TOPIC NAME : _____ DAY : Farria-50

TIME : DATE : / /

. Metal ion oxidation: Higher oxidation states of the metal ion increase the charge density on metal, leading to stronger ligand interactions and large Δ values favoring low-spin complexes.

(ii) Pairing energy:

Δ < P : Electrons will occupy the higher energy orbitals first, resulting in a high spin complex with more unpaired electron.

Δ > P : Electrons will pair up in the lower energy orbitals, resulting in a low-spin complex with fewer unpaired electron.

□ Why do the octahedral complexes show both high spin and low spin?

Ans:

In an octahedral complex, the spin of the complexe depends on the type of ligand. According to the spectrochemical series, the strong ligand cause a very strong repulsion wit metal d-orbitals. Thus $\Delta_o > P$ i.e. as pairing energy is low, it is thermodynamically more favourable to be paire than to move in higher energy state. Therefore stron ligand shows low spin.

Weaker ligands cause a very small splitting so that electron can easily move to higher energy state than to be paired [$\Delta_o < P$]. Thus they show high spin.

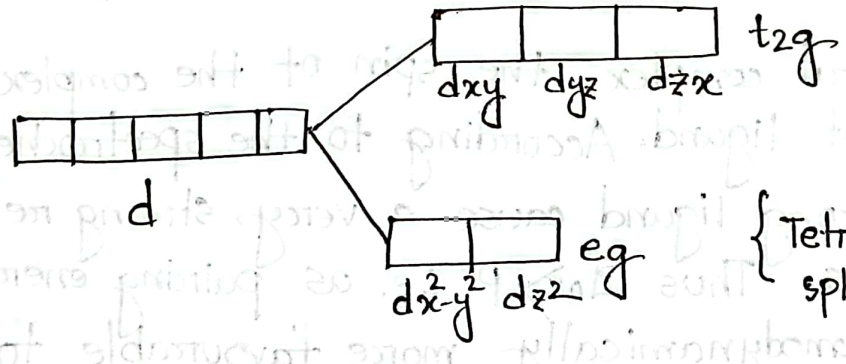
TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

□ Discuss the splitting of d-orbitals for Tetrahedral & square planar complex.

Ans: Tetrahedral:

The five degenerated d-orbitals split into two different energy levels when the metal ion is surrounded by the ligands. In a tetrahedral complex, the ligands approach along the corners of the tetrahedron; none of the d-orbitals point directly towards the ligand. As t_{2g}- d_{xy}, d_{yz}, d_{zx} are closer to the corners of the tetrahedron, they acquire the higher energy and orbitals along the axis with the least repulsion obtain lower energy. As, none of the d-orbitals can't point directly at the ligands, the splitting energy of the d-orbitals is also low.



{Tetrahedral + Octahedral splitting}

Ques:

□ Why the splitting pattern of d orbital is completely different from tetrahedral for octahedral complex?

□ Why the splitting energy is lower for tetrahedral complex than octahedral complex?

↳ যেহেতু octahedral ↳ along the axis ligand আসে তাই repulsion অনেক বেশি হয়, splitting energy বেশি। Tetrahedral এর ক্ষেত্রে সরাসরি d-orbital বরাবর ligand না আসায় repulsion কম। Octahedral ↳ 6 টি ligand থাকায় repulsion আরো বেশি হয়।

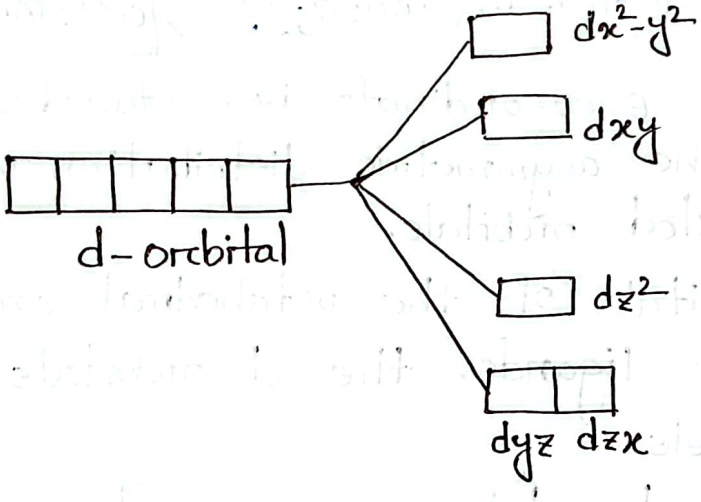
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TOPIC NAME : _____ DAY : Farria-50

TIME : DATE : / /

Square planar :

In a square planar complex, ligands directly point towards x and y axis. Thus dx^2-y^2 experiences the highest repulsive energy. dxy orbital lies in the plane of the ligand causing a significant repulsion. But no ligand approach along the z axis, due to it's geometry. Thus d-orbitals with z-component will fall in energy. The d^8 metal ions generally exhibit square planar geometry.



TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

□ What is Jahn-Teller Theorem ?

- John-Teller Theorem is a principle that states "Any non-linear molecular system in a degenerate electronic state will be unstable and will undergo distortion to form a system of lower energy and lower symmetry thereby removing the degeneracy."

□ On the basis of Jahn-Teller Theorem, explain the distortion of six co-ordinated complex. / Explain the Jahn-Teller distortion in $[Cu(H_2O)_6]^{2+}$ → $[Cu^{2+}: [Ar]\, 3d^9$, asymmetric

Ans:

The distortion of 6 co-ordinate i.e. octahedral complexes occurs due to the asymmetric distribution of electrons in the degenerated orbitals.

When the d-orbitals of the octahedral complex come in contact with ligands, the d-orbitals split into two energy levels:

T_{2g} → d_{xy}, d_{yz}, d_{zx} [Lower energy]

E_g → $d_{x^2-y^2}$, d_{z^2} [Higher energy]

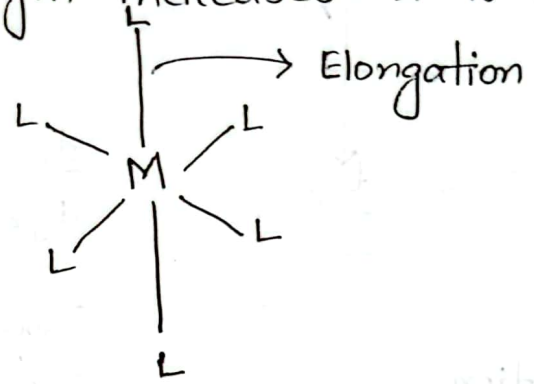
When there is an asymmetric distribution of electron in t_{2g} or e_g orbitals, it leads to a distortion of the complex. This results in a change of bond lengths. Generally high spin d^4 & d^9 also low spin d^7 exhibit strong distortion effect.

Two types of distortion occurs -

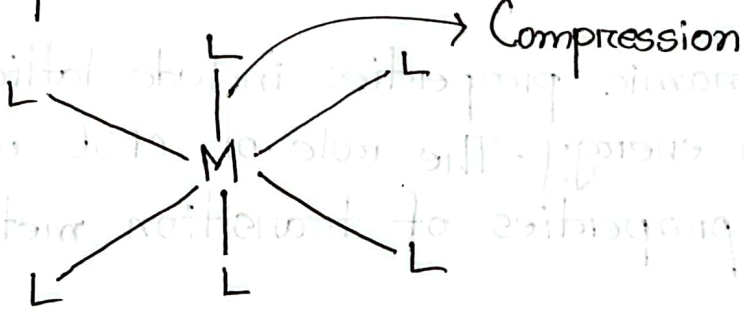
TOPIC NAME : _____ DAY : Faria-so

TIME : DATE : / /

(i) z - elongation: It occurs when a molecule or ion elongates along z-axis. If the asymmetric electron occupies dz^2, it can repulse the ligands from two sides and the axial bond length increases. It is called elongated octahedral.



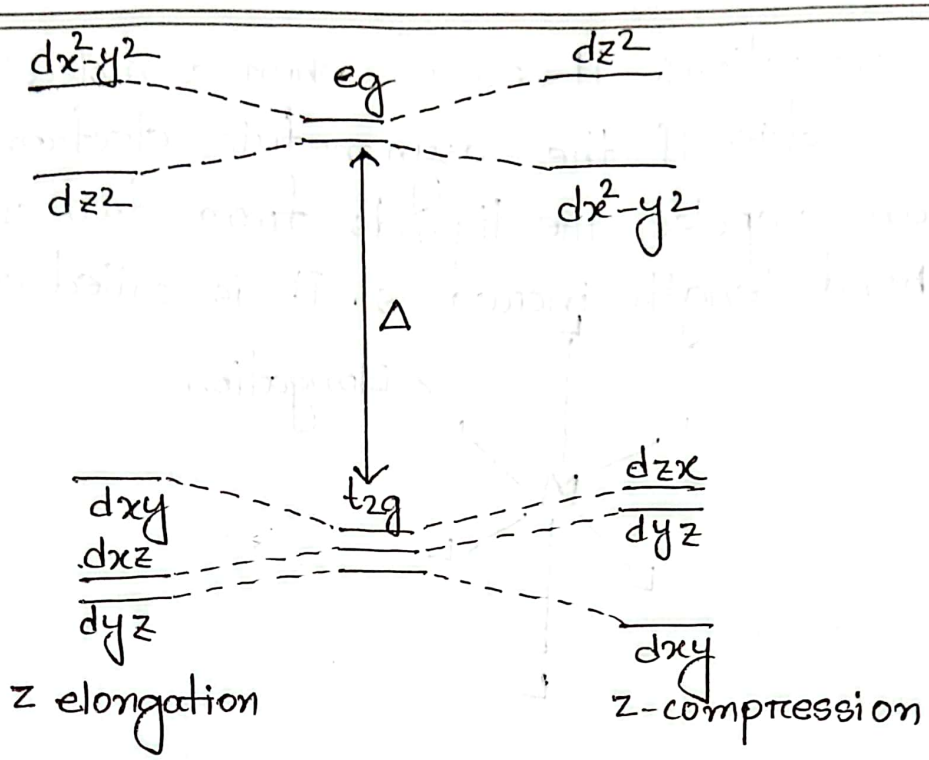
(ii) z - compression: It occurs when a molecule compresses along the z-axis. When ligand approaches along the axis orbital dx^2-y2 which contain asymmetric electron, it repulse them from four sides and the four equtorial bond length increases keeping other two short. This is called z-compression.



z-elongation causes the in lowering of energy those d-orbital that carry z-component; i.e. dz^2, dxz, dyz, where z-compression causes z-component to become higher in energy.

TOPIC NAME : _____ DAY : _____

TIME : DATE : / /



This Jahn-Teller distortions are strongly observed when asymmetric distribution of e^- occurs in e_g orbitals. The Jahn-Teller effect is weak when the asymmetry is in t_{2g} - orbitals.

□ Discuss the role of crystal field theory / CFSE upon the thermodynamic properties of transition metals.

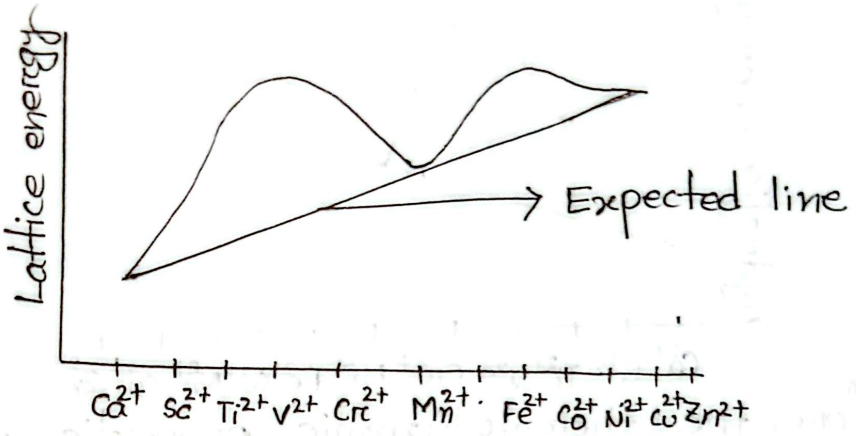
Ans:

Thermodynamic properties include lattice energy and hydration energy. The role of CFSE upon the thermodynamic properties of transition metals is given below:

(i) Lattice energy: It is the energy required to break apart an ionic solid and convert its constituent atoms into gaseous ions. The lattice enegy is expected to increase continuously accross the transition

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series as we move left to right due to the decrease in ionic radius of metal ions. But, expected straight line is not seen; instead a double hump shaped curve is obtained. This deviation is observed because of the CFE value of the metals.



Metals :	Ca^{2+}	Sc^{2+}	Ti^{2+}	V^{2+}	Cr^{2+}	Mn^{2+}	Fe^{2+}	Co^{2+}	Ni^{2+}	Cu^{2+}	Zn^{2}
Configuration:	d^0	d^1	d^2	d^3	d^4	d^5	d^6	d^7	d^8	d^9	d^{1}
CFSE	0	-0·4	-0·8	-1·2	-0·6	0	-0·4	-0·8	-1·2	-0·6	0

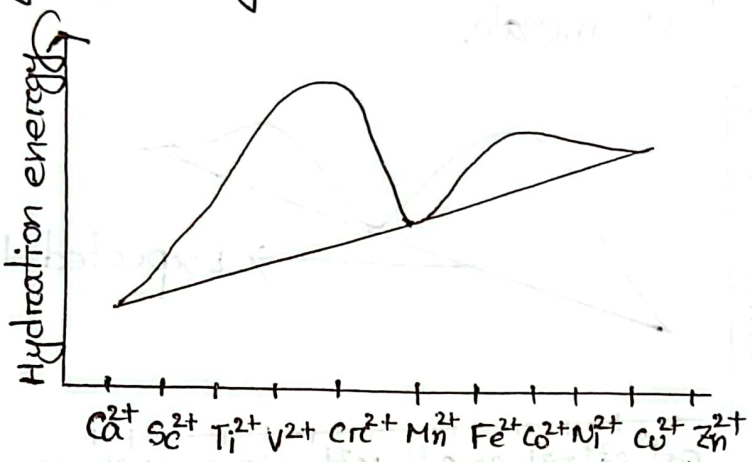
The CFE value of Ca^{2+}, Mn^{2+} & Zn^{2+} is 0. Hence they follow the expected line. But with the increase and decrease of the value, deviations of lattice energy from the expected line is observed. Hence we observe a double hump shaped graph. Therefore, the lattice energy of transition metals is highly affected by CFE values.

(ii) Heat of hydration: It is the energy released when new bonds between metal ions and water molecules are made. It is also expected to increase smoothly on going from left to right. But due to the differences in

TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

CSFE value, only Ca^{2+}, Mn^{2+} and Zn^{2+} follow expected line. Again same kind of increase and decrease is observed which produce two humped shaped graph again.



Therefore, the thermodynamic properties of transition metals depend on CSFE.

□ $[CrF_6]^{4-}$ & $[MnF_6]^{4-}$ both have six fluoride ligand. But all Mn-F bonds are equivalent whereas Cr-F are not. Explain.

Ans:

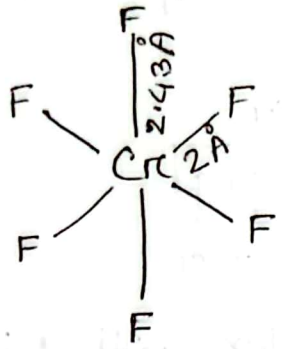
$[CrF_6]^{4-} \rightarrow Cr^{2+}$, $[MnF_6]^{4-} \rightarrow Mn^{2+}$

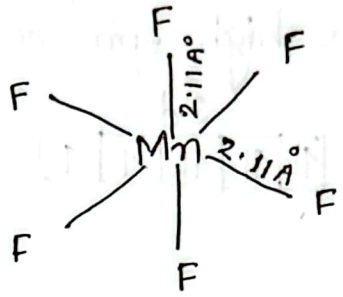
Cr^{2+}: $[Ar]\ 3d^4$ Cr^{2+} : $t_{2g}^3\ eg^1$

Mn^{2+}: $[Ar]\ 3d^5$ Mn^{2+} : $t_{2g}^3\ eg^2$

Here, Mn^{2+} has a symmetric distribution of electron in both t_{2g} and eg orbitals showing equal bond lengths. But in Cr^{2+}, e^- is not symmetrically distributed in eg orbital, which leads to Jahn-Teller distortion. The complex elongates or compress along z-axis causing

unequal Cr-F bond lengths.





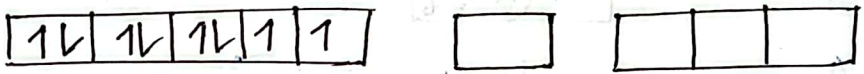
□ Ni^{2+} reacts with CN^-, Cl^- & NH_3. One of them is octahedral where others are square planar and tetrahedral. Assign ligands to corresponding complexes and give chemical formulae with suitable explanation.

Ans:

Ni^{2+} forms square planar complex with CN^-, tetrahedral with Cl^- and octahedral with NH_3.

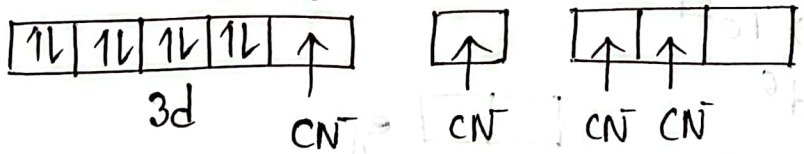
The e⁻ configuration of Ni^{2+} : $[Ar]\ 3d^8$

Ni^{2+} :



CN^- is a strong ligand which can pair the e⁻ of 3d orbital.

Ni^{2+} :



The hybridization is dsp^2. Hence CN^- forms square planar geometry with Ni^{2+}.

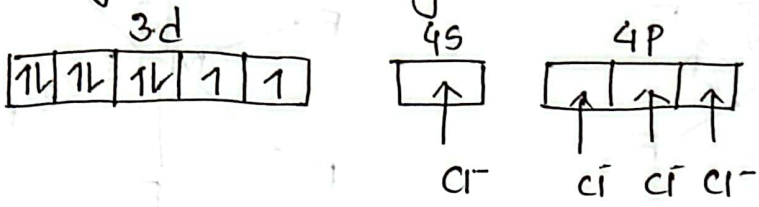
Chemical formula : $[Ni(CN)_4]^{2-}$

TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

Cl^- is a weak ligand which can't pair electron. Hence it gives high spin configuration.

Ni^{2+} :

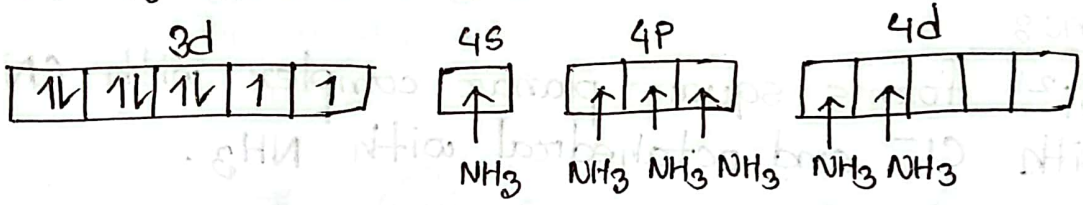


Thus Cl^- forms tetrahedral complex with Ni^{2+}.

Chemical formula : $[NiCl_4]^{2-}$

NH_3 has a lone pair electron & is neutral which allows 6 ligands to co-ordinate around Ni^{2+}.

Ni^{2+} :



Having six ligands, Ni^{2+} forms an octahedral geometry.

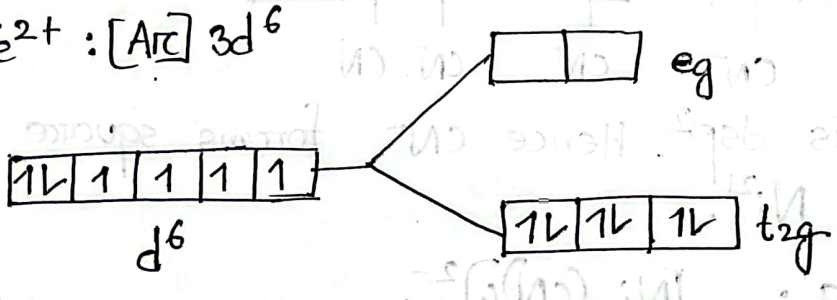
Chemical formula : $[Ni(NH_3)_6]^{2+}$

□ $[Fe(CN)_6]^{4-}$ is diamagnetic but $[FeF_6]^{3-}$ is paramagnetic - why?

Ans:

$[Fe(CN)_6]^{4-} \rightarrow Fe^{2+}$

$Fe^{2+} : [Ar]\ 3d^6$

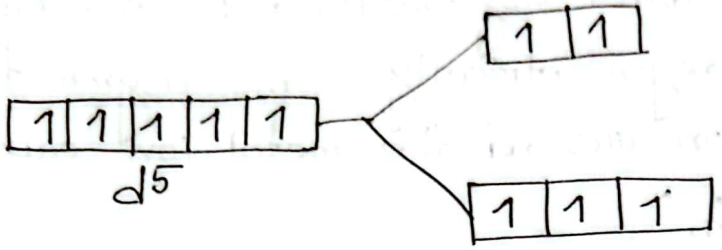


CN^- being ~~weak~~ strong field, shows low spin. There remains no unpaired electron, so it is diamagnetic.

TOPIC NAME : _____ DAY : Farria-50

TIME : DATE : / /

$[FeF_6]^{3-} \longrightarrow Fe^{3+} : 3d^5$



F^- being weak ligand shows high spin. Having five unpaired electrons, it is highly paramagnetic.

□ Of the first row transition metal complexes of formula $[M(NH_3)_6]^{3+}$ which metals are predicted by Jahn-Teller theorem to have distorted complexes.

Ans:

NH_3 is a low spin ligand.

Thus the configuration of the metals 3d orbital in $[M(NH_3)_6]^{3+}$:

$Sc^{3+} : [Ar]\ 3d^0$	$Sc^{3+} : t_{2g}^0\ eg^0$
$Ti^{3+} : [Ar]\ 3d^1$	$Ti^{3+} : t_{2g}^1\ eg^0$
$V^{3+} : [Ar]\ 3d^2$	$V^{3+} : t_{2g}^2\ eg^0$
$Cr^{3+} : [Ar]\ 3d^3$	$Cr^{3+} : t_{2g}^3\ eg^0$
$Mn^{3+} : [Ar]\ 3d^4$	$Mn^{3+} : t_{2g}^4\ eg^0$
$Fe^{3+} : [Ar]\ 3d^5$	$Fe^{3+} : t_{2g}^5\ eg^0$
$Co^{3+} : [Ar]\ 3d^6$	$Co^{3+} : t_{2g}^6\ eg^0$
$Ni^{3+} : [Ar]\ 3d^7$	$Ni^{3+} : t_{2g}^6\ eg^1$
$Cu^{3+} : [Ar]\ 3d^8$	$Cu^{3+} : t_{2g}^6\ eg^2$

[Zn & Sc is not transition metal]

TOPIC NAME : _____ DAY : _____ TIME : DATE : / /

From the configuration, it is seen that Ti^{3+}, V^{3+}, Mn^{3+} Fe^{3+} occupies t_{2g} orbital asymmetrically and Ni^{3+} occupies e_g orbital asymmetrically. This asymmetric distribution of electron around the metal ion causes Jahn-Teller distortion.

□ $TiCl_4$ is colorless whereas $[Ti(H_2O)_6]^{3+}$ is violet

Ans:

$TiCl_4$: $\rightarrow Ti^{4+}$

Ti : [Ar] $3d^2$ $4s^2$

Ti^{4+} : [Ar] $3d^0$

As in Ti^{4+} ion, there remains no electron in the 3d orbital, no d-d transition occurs. Thus it can't absorb or emit radiation and can't appear as colorful.

$[Ti(H_2O)_6]^{3+}$ → Violet [Discussed]

□ Draw the molecular orbital energy diagram of $[Ni(NH_3)_6]^{2+}$ and mention its magnetic nature.

Ans:

$[Ni(NH_3)_6]^{2+}$ → Ni^{2+}

Ni^{2+}: [Ar] $3d^8$, 12 electron by six NH_3.

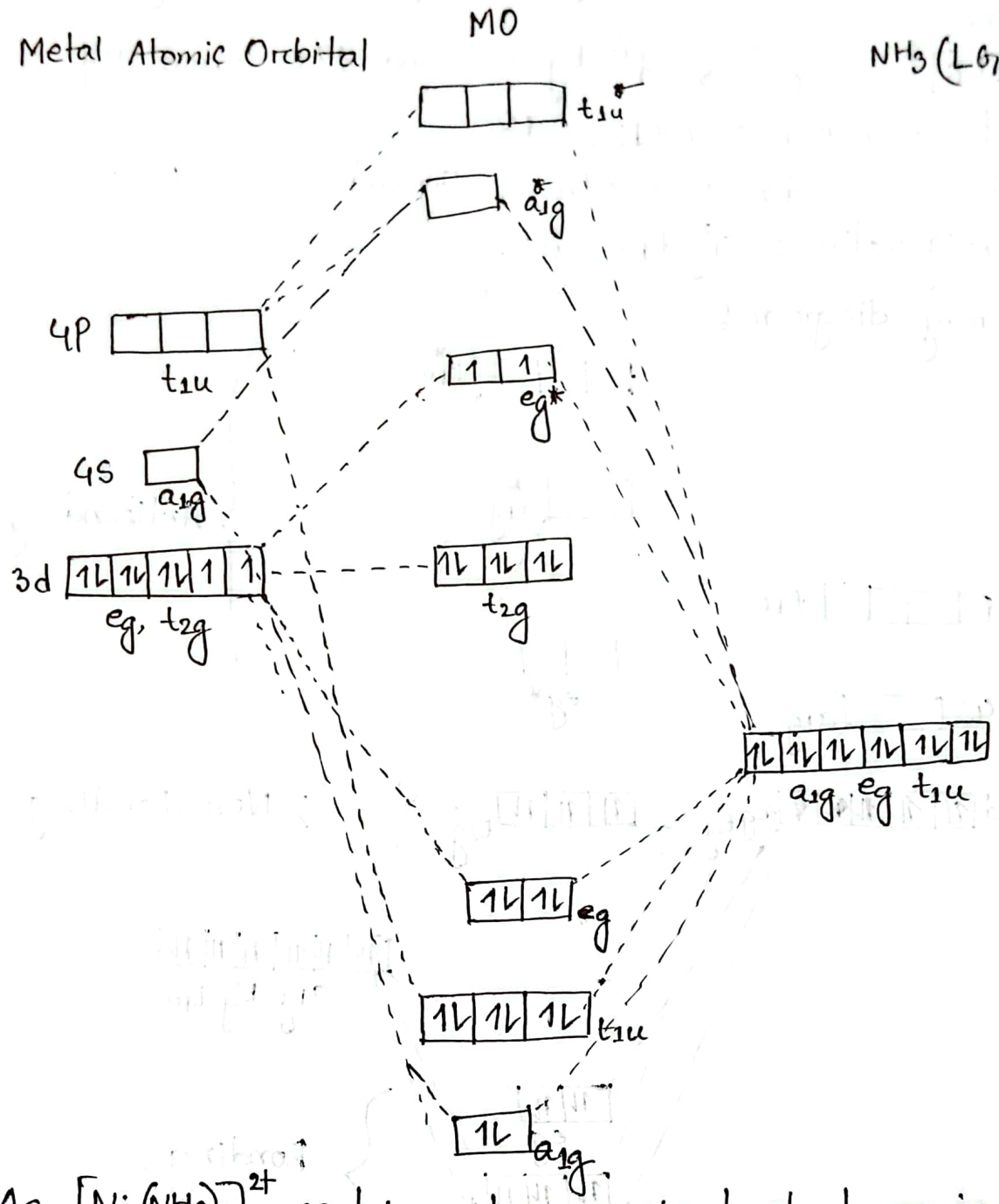
Total $e^- = 8 + 12 = 20$.

The electronic configuration is, $a_{1g}^2 t_{1u}^6 e_g^4 t_{2g}^6 e_g^{*2}$

TOPIC NAME : _____ DAY : Farria - 50

TIME : DATE : / /

For octahedral complex:



As $[Ni(NH_3)_6]^{2+}$ contains two unpaired electron in it's anti bonding orbital, it is paramagnetic.

TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

□ Draw a molecular orbital energy diagram for an octahedral complex $[CoL_6]^{3-}$ showing bonding, non-bonding and antibonding.

Ans

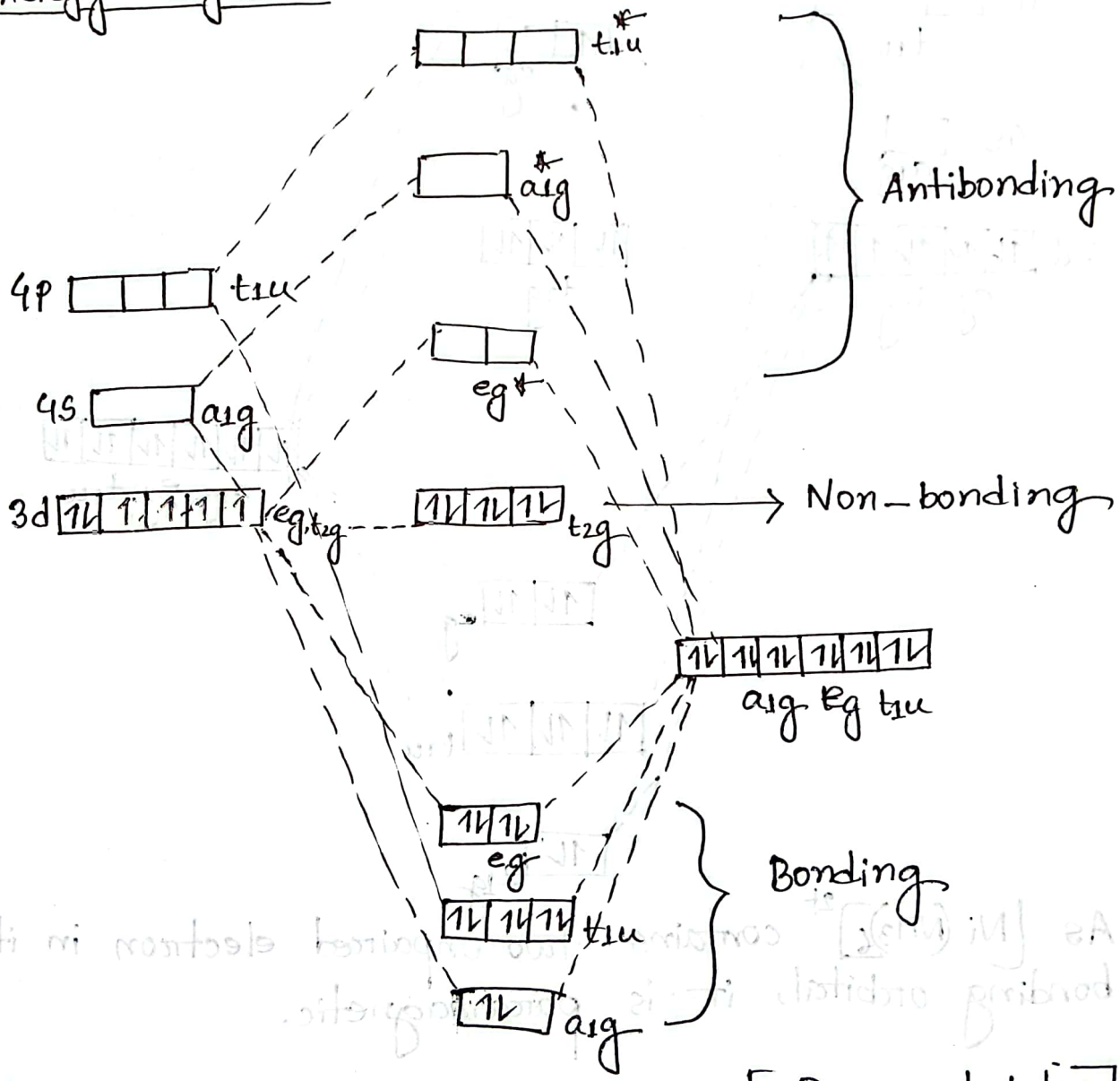
$[CoL_6]^{3-} \rightarrow Co^{3+} \rightarrow 3d^6$ by Co and $12e^-$ from ligand.

Total electron = $6+12 = 18e^-$

Let us consider, L = Strong ligand = Low spin.

Configuration: $a_{1g}^2\ t_{1u}^6\ e_g^4\ t_{2g}^6$

Energy diagram:



If L = weak ligand = High spin = [Discussed later]

TOPIC NAME : _____ DAY : Farria - 50

TIME : DATE : / /

Molecular Orbital Theory / Ligand Field Theory:

According to this theory, covalent bonds are formed between and ligand.

The representation of metal orbital in complex:

Octahedral Complex:

a → Non-degenerate
e → Double degenerate
t → Triple degenerate

s → g d → g
P → u f → u

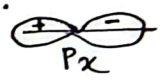
g = gerade u = ungerade

Metal orbital:

s → a_{1g}
P_x, P_y, P_z → t_{1u}
d_{xy}, d_{yz}, d_{zx} → t_{2g}
$d_{x^2-y^2}, d_{z^2}$ → e_g

অরবিটালের বিপরীত lobe এ same sign থাকলে gerade
" " " " opposite " " ungarde.





σ- bonding in Octahedral complex:

Ligand Group Orbitals (LGO):

(1) As ligand in a octahedral complex approaches along the axis, the orbitals which align about axis may bond with ligands.

(i) a_{1g} (s orbital), t_{1u} (P_x, P_y, P_z), e_g ($d_{x^2-y^2}, d_{z^2}$) orbital are along the axis. So, they significantly overlap with ligand

(ii) But t_{2g} (d_{xy}, d_{yz}, d_{zx}) lies between the axis, doesn't overlap with ligand and remains as non bonding.

TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

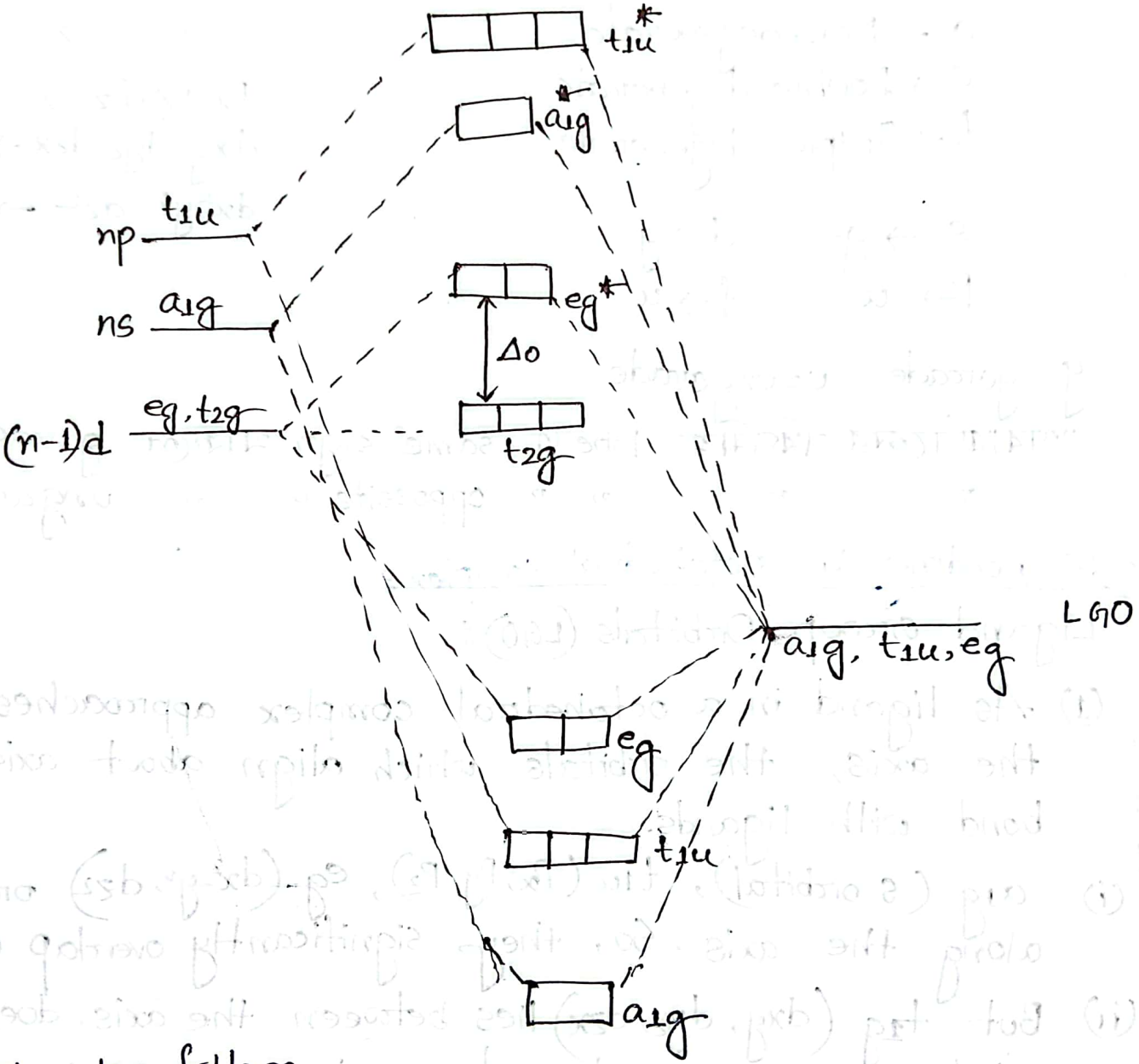
Energy order,

$a_{1g} < t_{1u} < e_g < t_{2g} < e_g^* < a_{1g}^* < t_{1u}^*$

The extent of overlap of,

metal a_{1g} & LGO a_{1g} > Metal t_{1u} and LGO t_{1u} > metal e_g & LGO e_g.

e- গুলো এই order ও অরবিটাল পুর্ন করবে।



It also follows,

$\Delta_o > P \rightarrow$ pairing

$\Delta_o < P \rightarrow$ High spin

TOPIC NAME : _____ DAY : Farria - 50

TIME : DATE : / /

Example:

$[Co(NH_3)_6]^{3+} \rightarrow Co^{3+} \rightarrow 3d^6$ by metal, $12e^-$ from NH_3

Total electron = 6+12 = 18.

NH_3 is a strong ligand shows low spin.

Configuration: $a_{1g}^2\ t_{1u}^6\ eg^4\ t_{2g}^6$. $[\Delta_o > P]$

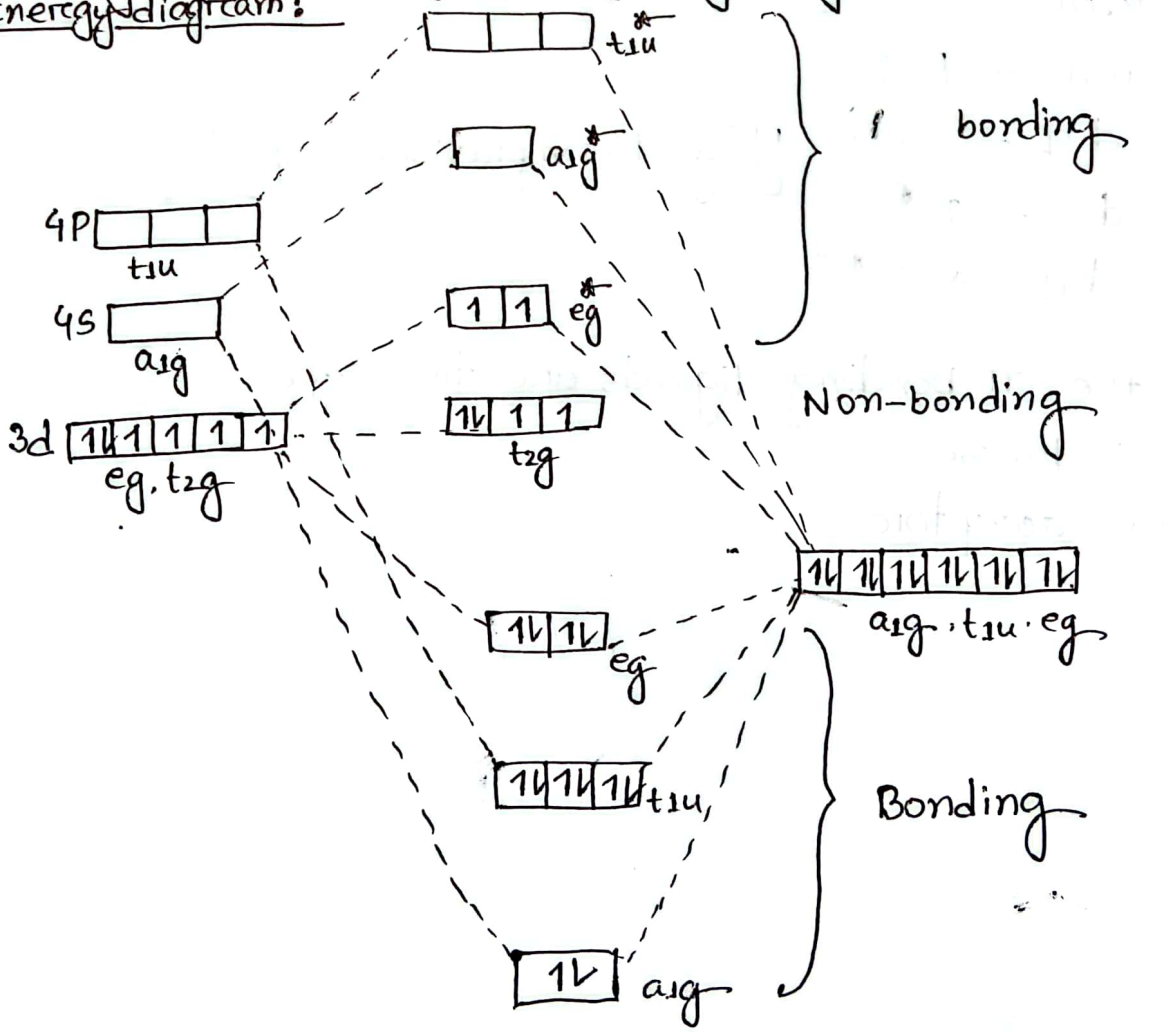
$[CoF_6]^{3-} \rightarrow Co^{3+} \rightarrow d^6$, $12e^-$ from F^-

Total electron = 6+12 = 18

F^- is a weak ligand shows high spin.

Configuration: $a_{1g}^2\ t_{1u}^6\ eg^4\ t_{2g}^4\ eg^{2*}$ $[\Delta_o < P]$

Energy diagram:



TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

π - bonding in octahedral Complex :

- In case of LGO and metal AO contain t_{2g} and t_{1u} symmetry can participate in the sidewise overlaping on π bonding in octahedral geometry.
- But the orbital already participating in σ bonding (t_{1u}) do not show π bonding. Thus only t_{2g} shows π bonding.
- There are 4 types of LGO that have same as t_{2g} symmetry ; $p\pi$, $d\pi$, π^*, σ^*

4 possible interactions :

MAO	LGOS
$d\pi$	$p\pi$
$d\pi$	$d\pi'$
$d\pi$	π^*
$d\pi$	σ^*

[Fig From previous chapter]

In the π bonding ligands are two types :

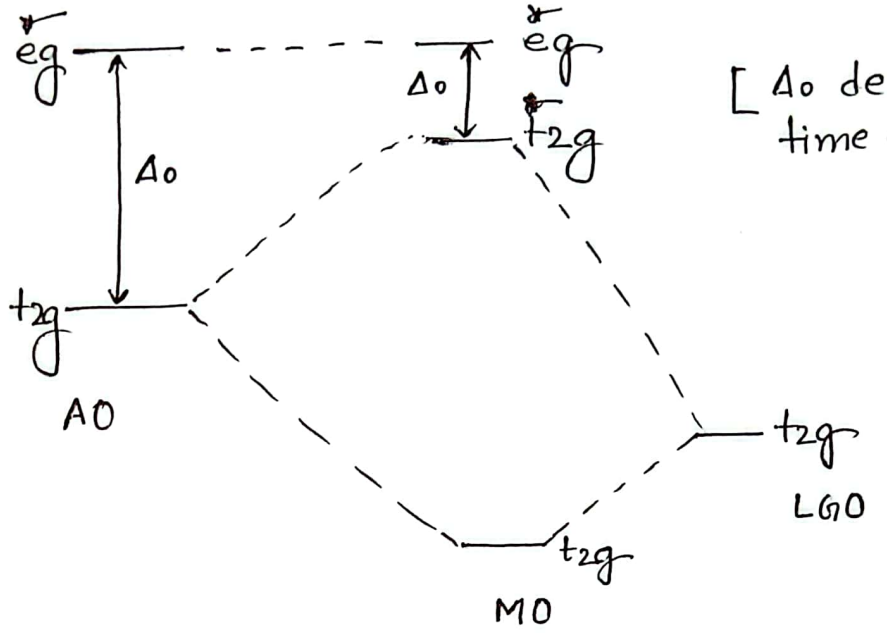
(i) π donor

(ii) π acceptor

TOPIC NAME : _____ DAY : Faria - 50

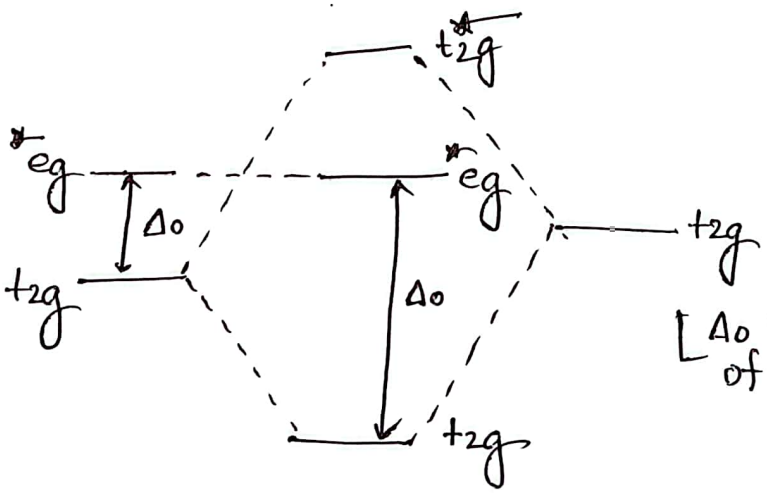
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π - donors:



[Δo decreases as the time of π bonding]

π - acceptors:



[Δo increases as the time of π bonding]